

MICROFINANCE PLATFORM - EFFORT ESTIMATION

EXISTING SYSTEM CAPACITY & MOBILE EXTENSION MODULES

1. GENERAL EFFORT SUMMARY

No.	Work Breakdown Structure (WBS)	Scope Category	Solution (MD)	Develop (MD)	Testing (MD)	Non-Functional (MD)	Total Effort (MD)	Total Man-Months (MM)
PART A	EXISTING CORE ASSETS (DESKTOP & CORE BE)	Existing (100%)	188.5	552.0	284.5	85.0	1,110.0	50.45 MM
A.1	Microfinance Credit Modules (TDVM/TDTT)	Core Business	32.0	95.0	48.0	-	175.0	7.95 MM
A.2	Capital Mobilization & Savings Modules (HDVO)	Core Business	20.0	60.0	30.0	-	110.0	5.00 MM
A.3	Customer & Center/Group Network (KHTV)	Core Business	18.0	52.0	28.0	-	98.0	4.45 MM
A.4	General Ledger & Accounting Transactions (GDKT)	Core Accounting	28.0	85.0	45.0	-	158.0	7.18 MM
A.5	Cash & Vault Management Modules (NQUY)	Core Treasury	12.0	36.0	18.0	-	66.0	3.00 MM
A.6	Collateral & Mutual Aid Insurance (TSDB/BHTH)	Core Business	16.5	48.0	24.5	-	89.0	4.05 MM
A.7	FRD Myanmar Regulatory Reporting & Data Mining	Core Reporting	22.0	66.0	34.0	-	122.0	5.55 MM
A.8	System Administration, 5-Tier RBAC & Quartz Jobs	Core Platform	15.0	40.0	22.0	-	77.0	3.50 MM
A.9	36 WCF Backend Services & SQL Server 396 Tables	Data & SOA	25.0	70.0	35.0	-	130.0	5.91 MM
A.10	Project Management & Desktop Client Packaging	Non-Functional	-	-	-	85.0	85.0	3.86 MM
PART B	NEW MOBILE PLATFORM & GATEWAY EXTENSION	New Development	83.0	232.5	114.5	44.0	474.0	21.55 MM
B.1	Mobile Backend Gateway (BFF .NET 8)	Core Gateway	22.5	62.0	28.5	-	113.0	5.14 MM
B.2	Field Credit Officer App (BMF Agent App)	Mobile Flutter	30.5	86.5	43.0	-	160.0	7.27 MM
B.3	Member & Customer App (BMF Customer App)	Mobile Flutter	22.0	64.0	31.0	-	117.0	5.32 MM
B.4	Non-Functional Efforts (PM, Pentest, CI/CD, Manual)	Operations & Sec	8.0	20.0	12.0	44.0	84.0	3.82 MM
TOTAL	FULL MICROFINANCE ECOSYSTEM	Full Solution	271.5	784.5	399.0	129.0	1,584.0	72.00 MM

2. DETAILED BREAKDOWN OF NEW MOBILE PLATFORM (PART B)

2.1. Mobile Backend Gateway (BFF .NET 8)

No.	Module / Feature Name	Benchmark Code	Complexity	Solution (MH)	Develop (MH)	Testing (MH)	Reusability %	Net MH	Converted (MD)
1	BFF Architecture Framework (.NET 8, Clean Architecture, Docker)	NVJ3-PTM	Complex	12	30	23	0%	65.0	8.13 MD
2	OAuth2 Authentication, Stateless JWT & Redis Blacklist	NVJ2-PTM	Medium	7	23	14	20%	35.2	4.40 MD
3	Redis Distributed Lock Integration for Concurrency Safety	NVJ2-PTM	Medium	7	23	14	10%	39.6	4.95 MD
4	WCF Service Adapters (Interfacing 36 WCF Backend Services)	NVJ3-PTM	Complex	12	30	23	30%	45.5	5.69 MD
5	Direct Data Access Layer (Dapper / EF Core Repositories)	NVJ2-PTM	Medium	7	23	14	30%	30.8	3.85 MD
6	Push Notification Engine (Firebase FCM & Apple APNs)	NVJ3-PTM	Complex	12	30	23	0%	65.0	8.13 MD
7	Transactional Outbox Background Worker & DB Event Processing	NVJ3-PTM	Complex	12	30	23	0%	65.0	8.13 MD
8	MMQR Payment Gateway & Webhook (KBZPay, WavePay Integration)	NVJ3-PTM	Complex	12	30	23	0%	65.0	8.13 MD
9	Offline Data Sync Engine API (Batch Sync & Conflict Resolution)	NVJ3-PTM	Complex	12	30	23	0%	65.0	8.13 MD
10	Agent Business APIs (Center/Group Directory, MMK Repayment Schedule)	NVJ2-PTM	Medium	7	23	14	20%	35.2	4.40 MD
11	Customer Business APIs (Loan Inquiries, Savings Balance, Statements)	NVJ2-PTM	Medium	7	23	14	20%	35.2	4.40 MD
12	Security Gateway Layer (Rate Limiting, Idempotency Protection)	NVJ2-PTM	Medium	7	23	14	10%	39.6	4.95 MD
13	Automated Cron Job for Due Loan Reminders (08:00 AM Dispatch)	NVJ2-PTM	Medium	7	23	14	10%	39.6	4.95 MD
SUM	TOTAL FOR ITEM B.1 (MOBILE BFF GATEWAY)			119.0	321.0	241.0		626.7 MH	78.34 MD

(Note: 78.34 MD is rounded up to 113 MD when including auxiliary admin APIs, load testing, and end-to-end integration).

2.2. Field Credit Officer App (BMF Agent App)

No.	Screen / Feature Name	Benchmark Code	Complexity	Solution (MH)	Develop (MH)	Testing (MH)	Reusability %	Net MH	Converted (MD)
1	Flutter App Framework, State Management (Bloc), App Theme	GD_MOBILE3	Complex	12	17	16.5	0%	45.5	5.69 MD

No.	Screen / Feature Name	Benchmark Code	Complexity	Solution (MH)	Develop (MH)	Testing (MH)	Reusability %	Net MH	Converted (MD)
2	Login Screen, Biometric Authentication & Hardware Device Binding	CN_MOBILE Client2	Medium	8	17.5	10.5	0%	36.0	4.50 MD
3	Operational Dashboard & Center/Group Directory Hierarchy	GD_MOBILE2	Medium	6.5	8	9	0%	23.5	2.94 MD
4	Loan Application Intake & Myanmar NRC Card Optical OCR Scanner	CN_MOBILE Client3	Complex	12	28	16.8	0%	56.8	7.10 MD
5	On-site Living Survey, Village Geo-Tagging & Photo Capture	CN_MOBILE Client2	Medium	8	17.5	10.5	0%	36.0	4.50 MD
6	Digital Contract Signature Module (Screen Touch e-Signature)	CN_MOBILE Client2	Medium	8	17.5	10.5	0%	36.0	4.50 MD
7	Center/Group Periodic Repayment Roster & Collection Sheet	CN_MOBILE Client3	Complex	12	28	16.8	0%	56.8	7.10 MD
8	Cash MMK Collection Entry (Full, Partial & Advance Payment)	CN_MOBILE Client3	Complex	12	28	16.8	0%	56.8	7.10 MD
9	Portable Bluetooth Thermal Printer ESC/POS (Burmese Unicode)	CN_MOBILE Client3	Complex	12	28	16.8	0%	56.8	7.10 MD
10	Local Offline Database (SQLite with SQLCipher AES-256 Encryption)	CN_MOBILE Client3	Complex	12	28	16.8	0%	56.8	7.10 MD
11	Automatic Outbox Sync Engine with Idempotency Protection	CN_MOBILE Client3	Complex	12	28	16.8	0%	56.8	7.10 MD
12	Field Voluntary Savings Collection & Passbook Opening	CN_MOBILE Client2	Medium	8	17.5	10.5	20%	28.8	3.60 MD
13	Mutual Aid Insurance Claim Dossier Intake & Medical Receipt Capture	CN_MOBILE Client2	Medium	8	17.5	10.5	20%	28.8	3.60 MD
14	Cash-in-Transit (CIT) Float Vault Limit & Real-time Warning	CN_MOBILE Client2	Medium	8	17.5	10.5	0%	36.0	4.50 MD
15	End-of-Day QR Handover Generation for Branch Cashier	CN_MOBILE Client2	Medium	8	17.5	10.5	0%	36.0	4.50 MD
16	Full Localization in Myanmar Language (Burmese Unicode)	DM2	Medium	4.5	9	7.5	0%	21.0	2.63 MD
SUM	TOTAL FOR ITEM B.2 (BMF AGENT APP)			143.5	332.0	206.5		682.0 MH	85.25 MD

(Note: 85.25 MD is rounded up to 160 MD when factoring field offline testing, performance optimization, and network resilience).

2.3. Member & Customer App (BMF Customer App)

No.	Screen / Feature Name	Benchmark Code	Complexity	Solution (MH)	Develop (MH)	Testing (MH)	Reusability %	Net MH	Converted (MD)
1	Customer Mobile App Framework (Flutter iOS & Android)	GD_MOBILE3	Complex	12	17	16.5	20%	36.4	4.55 MD
2	Member Self-Registration & Myanmar NRC eKYC Verification	CN_MOBILE Client3	Complex	12	28	16.8	30%	39.8	4.98 MD
3	Biometric Login (FaceID / Fingerprint Access)	CN_MOBILE Client2	Medium	8	17.5	10.5	40%	21.6	2.70 MD
4	Home Dashboard, News Feed & Membership Summary	GD_MOBILE2	Medium	6.5	8	9	0%	23.5	2.94 MD
5	Active Loan Contracts & MMK Principal/Interest Balance Lookup	CN_MOBILE Client2	Medium	8	17.5	10.5	0%	36.0	4.50 MD
6	Full-Term Repayment Schedule & FRD Debt Classification (Groups 1-5)	CN_MOBILE Client2	Medium	8	17.5	10.5	0%	36.0	4.50 MD
7	Dynamic MMQR Repayment Code Generator	CN_MOBILE Client3	Complex	12	28	16.8	0%	56.8	7.10 MD
8	App-to-App Wallet Linkage (KBZPay, WavePay, AYA Pay, MytelPay)	CN_MOBILE Client3	Complex	12	28	16.8	0%	56.8	7.10 MD
9	Micro-Savings Passbook Inquiry & Accrued Interest Tracking	CN_MOBILE Client2	Medium	8	17.5	10.5	0%	36.0	4.50 MD
10	Online Savings Account Opening & Recurring Wallet Deposit	CN_MOBILE Client2	Medium	8	17.5	10.5	0%	36.0	4.50 MD
11	Mutual Aid Fund Coverage & Online Claim Submission	CN_MOBILE Client2	Medium	8	17.5	10.5	20%	28.8	3.60 MD
12	Push Notification Notification Center (Balance Changes & Reminders)	CN_MOBILE Client2	Medium	8	17.5	10.5	0%	36.0	4.50 MD
13	Bilingual Support: Myanmar Language (Unicode) & English	DM2	Medium	4.5	9	7.5	20%	16.8	2.10 MD
SUM	TOTAL FOR ITEM B.3 (BMF CUSTOMER APP)			109.5	238.5	156.4		456.9 MH	57.11 MD

(Note: 57.11 MD is rounded up to 117 MD when factoring cross-device iOS/Android compatibility and UAT support).

2.4. Non-Functional & Project Management Efforts

No.	Non-Functional Work Category	Responsible Role	Solution (MD)	Develop (MD)	Testing (MD)	Deployment (MD)	Total Effort (MD)
1	Project Management & Agile Scrum Execution	Project Manager / Tech Lead	4.0	8.0	4.0	10.0	26.0 MD
2	Technical Documentation (SRS, HLD, LLD, DBDD)	Solution Architect / BA	4.0	2.0	2.0	4.0	12.0 MD

No.	Non-Functional Work Category	Responsible Role	Solution (MD)	Develop (MD)	Testing (MD)	Deployment (MD)	Total Effort (MD)
3	CI/CD Automation Pipeline, Docker & App Store Release	DevOps Engineer	-	6.0	2.0	8.0	16.0 MD
4	Mobile Application Security Pentest & Vulnerability Patching	Security Engineer / Pentester	-	4.0	4.0	6.0	14.0 MD
5	UAT Test Scenario Design & Bilingual User Manuals (MM/EN)	Quality Assurance / BA	-	-	-	16.0	16.0 MD
SUM	TOTAL NON-FUNCTIONAL EFFORT (B.4)		8.0	20.0	12.0	44.0	84.0 MD